

Do Function Vectors and Task Vectors Generalize?

A Reproducibility Study Across Model Families and Scale

Serendeeep Rudraraju University of Marburg rudraraj@students.uni-marburg.de
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Summary: where each vector beats its control

On word-pair tasks, mostly yes. On sentiment with arbitrary labels, task vectors stay near chance on every checkpoint.

	Test 1 Exploratory grid	Test 2 Held-out layers	Test 3 Sentiment
method, model	5–16 word-pair tasks per model. Layer chosen on the evaluation data. Share of the ICL gap recovered.	8 tasks. Each vector and each control picks its layer on dev data. Points above control.	SST-2-derived reviews, arbitrary labels (A/B and B/A). TV test accuracy; chance is 50%.
FV GPT-J-6B	✔ 0.50 control −0.01	✔ +35.9 CI 18.5 to 52.6	⊖ not run
TV GPT-J-6B	✔ 0.56 shuffled 0.28, $d = 0.78$	✔ +16.1 CI 4.1 to 30.3	⊖ 52.7% ICL 73.5%
FV Llama-3.1-8B	✔ 0.42 control 0.00	✔ +24.3 CI 7.6 to 42.9	⊖ not run
TV Llama-3.1-8B	✔ 0.83 shuffled 0.42	⊖ −4.2 CI −20.7 to 8.7	⊖ 50.0% ICL 90.8%
TV Llama-3.1-70B	✔ 0.80 shuffled 0.48	⊖ not run	⊖ not run
FV Gemma-2-9b base	⊖ not run	✖ 0% acc. 2 tasks, ICL 70–99%	⊖ not run
TV Gemma-2-9b base	⊖ not run	✔ +19.1 CI 3.9 to 35.3	⊖ 49.6% ICL 94.0%
FV Gemma-2-9b-it	✖ 0.01 control 0.01	✖ 0% acc. 2 tasks, ICL 68–91%	⊖ not run
TV Gemma-2-9b-it	✔ 0.83 shuffled 0.39	✔ +32.3 CI 7.4 to 57.9	⊖ 50.0% ICL 93.3%

Table 1: Tests 1→2 tighten layer selection; Test 3 changes the task. ✔ pass (Test 1) or CI above 0 (Test 2) ⚡ weak pass ($0.5 \leq d < 0.8$) ⊖ CI includes 0 ⊖ near chance ✖ fail or 0% accuracy ⊖ not run
Gemma cells in Tests 2 and 3 ran on Kaggle; all other runs used NDIF.

Background

Function vectors (FVs)^[1] add the summed outputs of causally selected attention heads to the residual stream. Task vectors (TVs)^[2] copy one hidden state from a demonstration prompt into a zero-shot pass. Both papers claim the vector carries an in-context task into a new prompt.

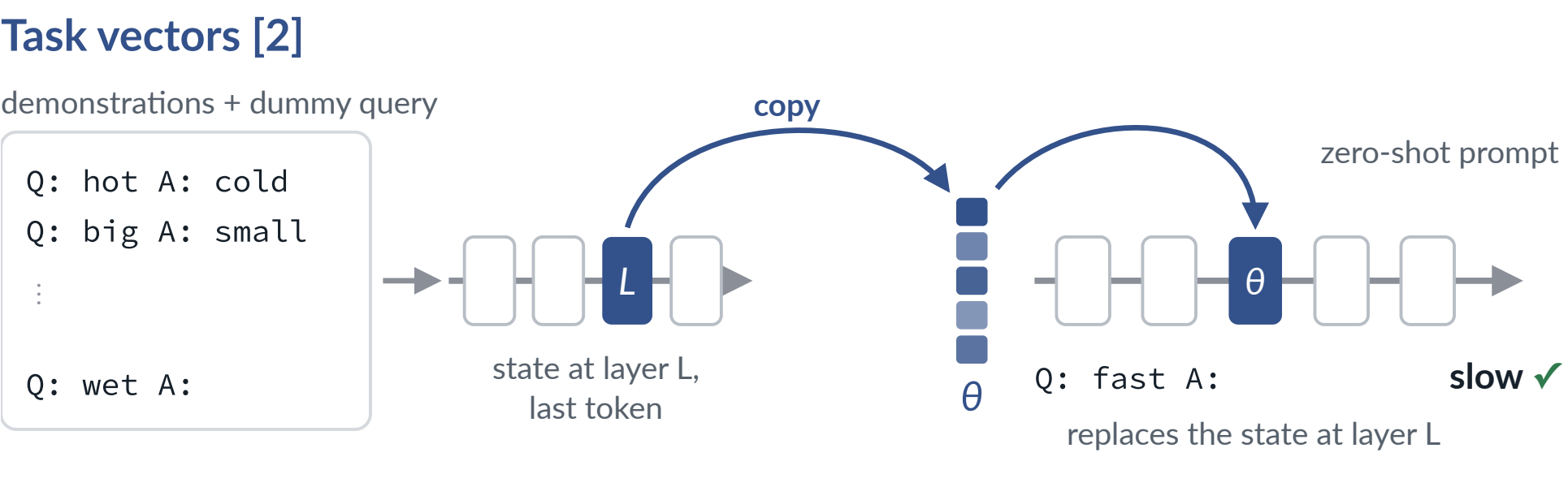
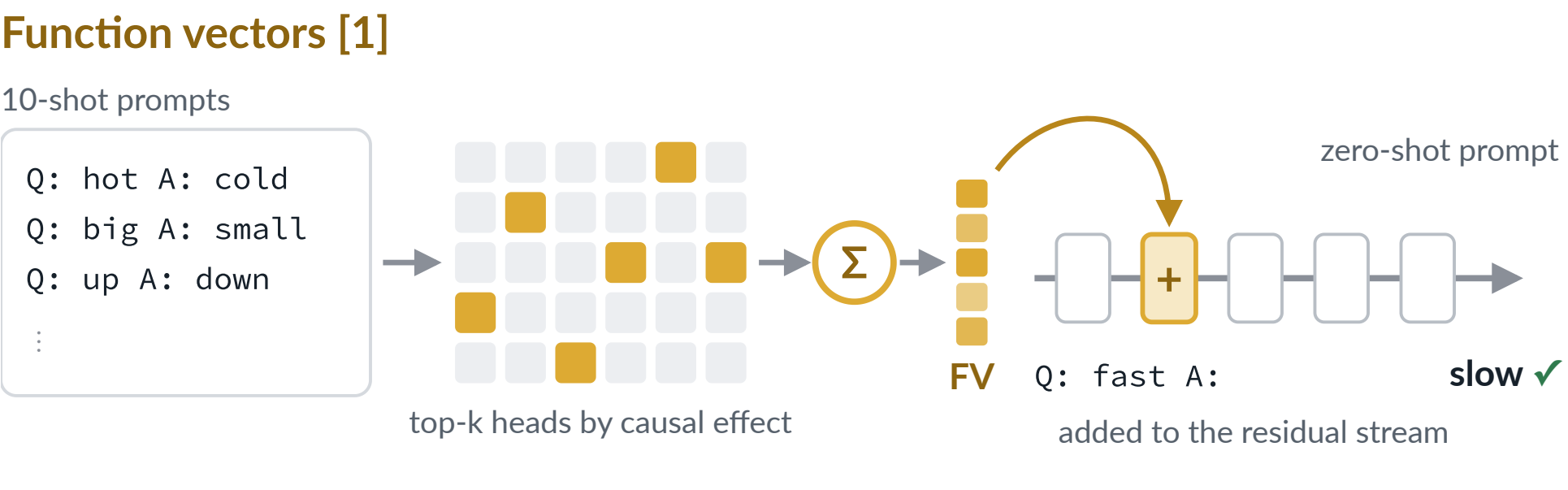


Figure 1: The two interventions, reimplemented in NNsight^[3]; the Kaggle Gemma runs use PyTorch hooks.

Setup

- GPT-J-6B (used by both papers), Llama-3.1-8B and 70B, and Gemma-2-9b in base and instruction-tuned versions.
- 16 word-pair tasks from [1], scored by top-1 first-token accuracy.
- Random vectors and random heads as FV controls; label-shuffled demonstrations as TV controls.

Test 1: Exploratory grid

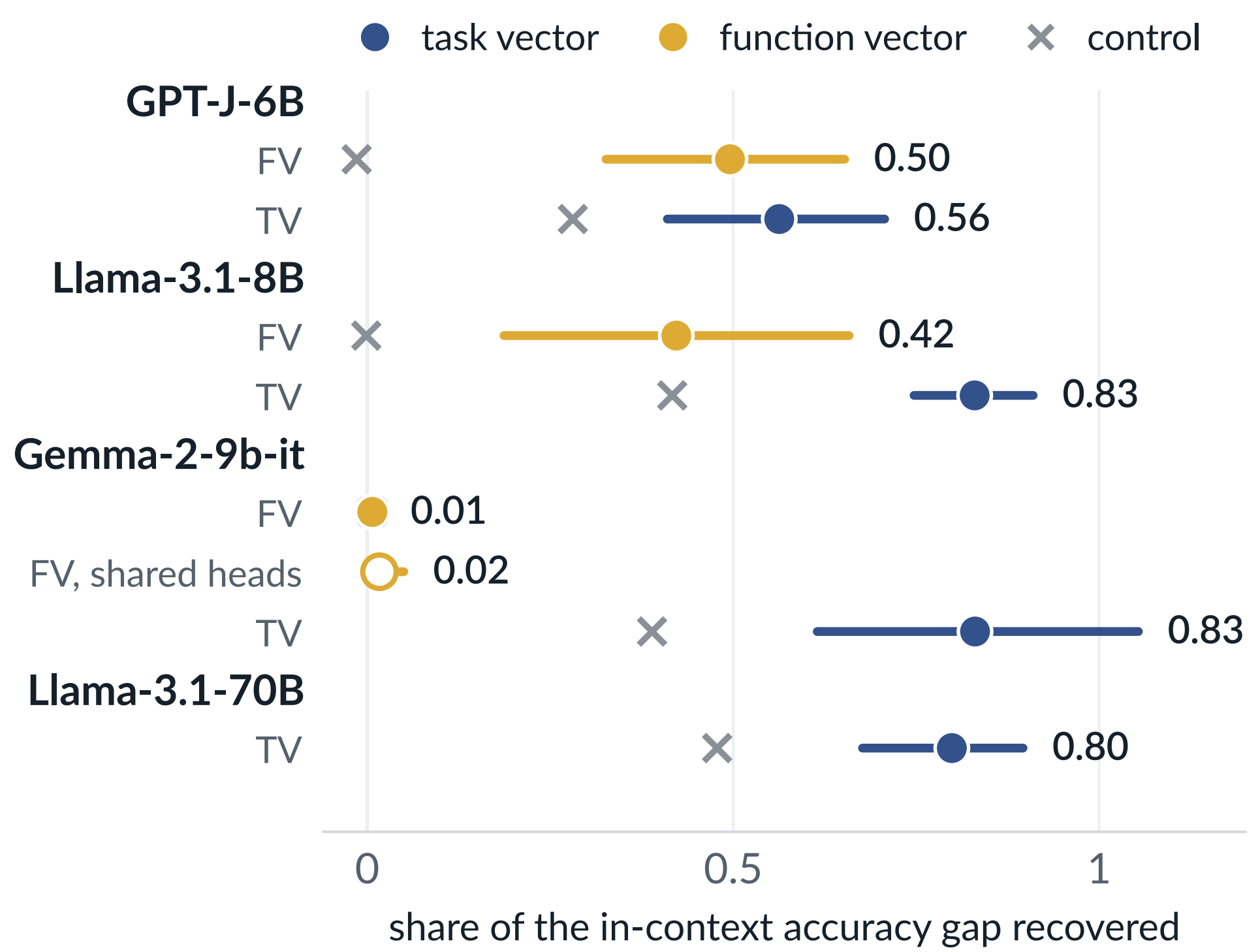


Figure 2: Best-layer recovery with task-bootstrap 95% CIs; × marks the control. Choosing the layer on the scored data favours the real vector.

The Gemma-2-9b-it FV recovers 0.01. Scaling it 8x, injecting before the post-attention RMSNorm, or picking up to 40 heads by cross-task AIE as in [1] does not restore a consistent advantage.

Test 2: Held-out layer selection

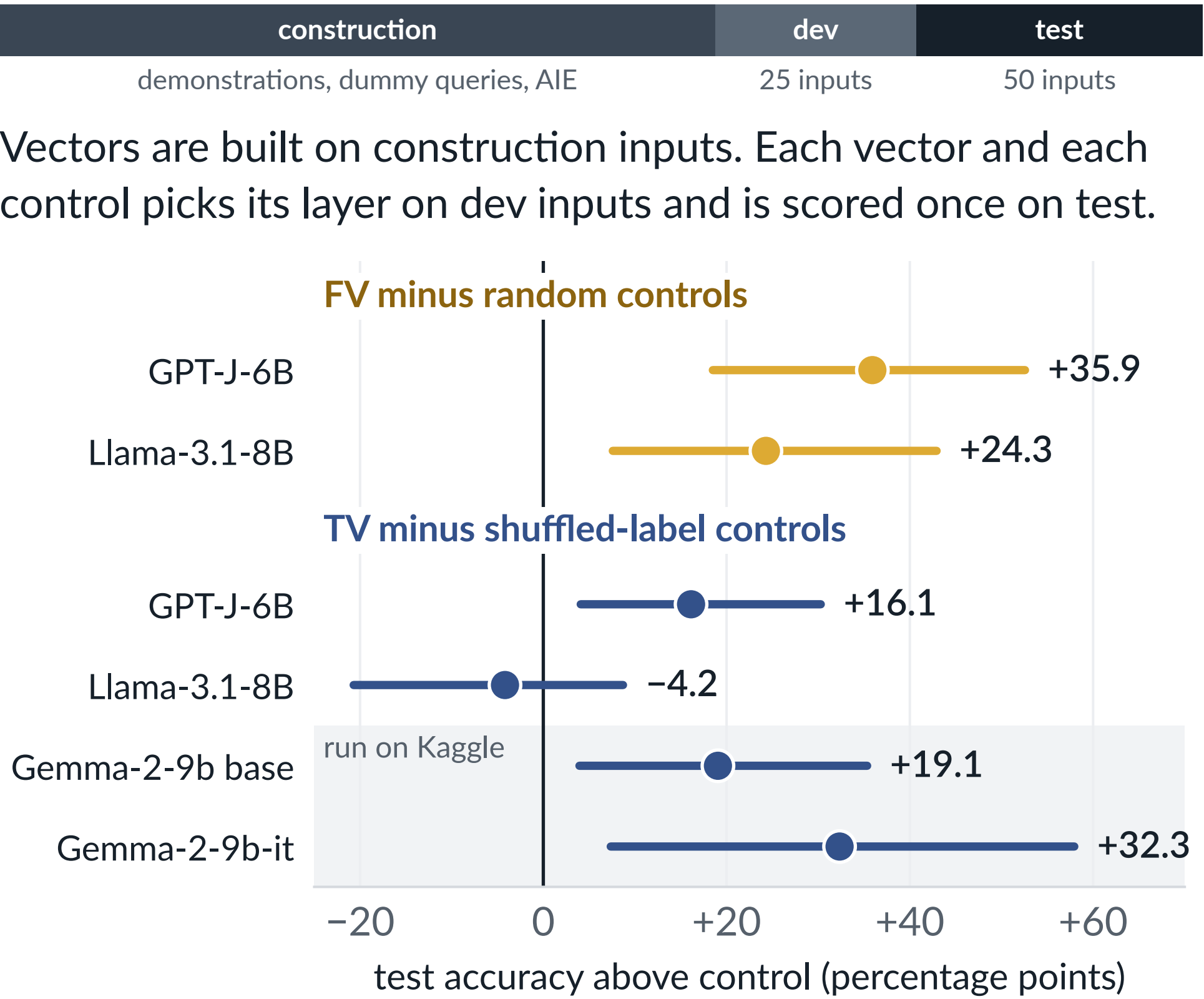


Figure 3: Test accuracy above control in percentage points; 95% CIs over eight task means. The gap between the two Gemma checkpoints is uncertain (13.3 points, CI −2.9 to 27.9).

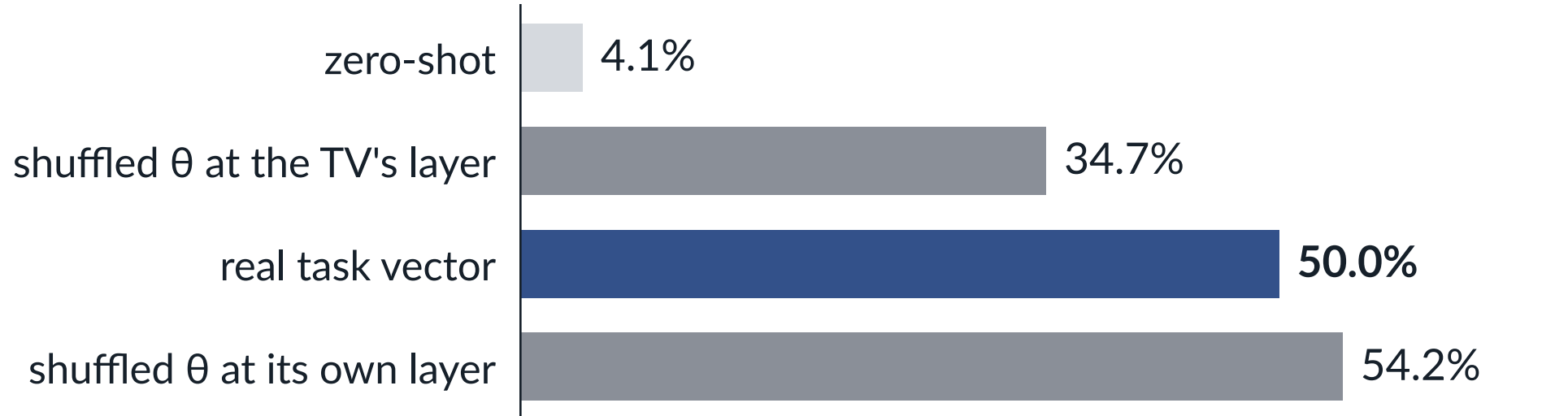


Figure 4: Llama-3.1-8B held-out test accuracy.

At the TV's layer, a shuffled-label vector scores 34.7%. At its own layer it scores 54.2%, against 50.0% for the real TV, so the Llama-3.1-8B comparison is inconclusive.

Gemma: base vs instruction-tuned

Both checkpoints share partitions and demonstrations. Kaggle and NDIF agree on 197 of 200 predictions in a limited check (Gemma IT, antonym).

FV, test accuracy (%)	ICL	FV	control
base, antonym	70.0	0.0	0.0
base, country-capital	98.7	0.0	0.0
IT, antonym	68.0	0.0	0.0
IT, country-capital	90.7	0.0	0.0

On these two tasks the base checkpoint fails too, so instruction tuning alone does not explain our FV null.

Limitations

- Our FV recipe picks 10 heads per task and skips the ICL-correct filtering of [1]; the Gemma null is about this implementation.
- Shuffled controls can keep some correct pairs.
- The Gemma TV and sentiment extensions were specified after earlier results. First-token accuracy only.

Test 3: Sentiment with arbitrary labels

Positive reviews map to A and negative to B, or the reverse, so label meaning gives no help^[4]. ICL learns the mapping. The TV does not.

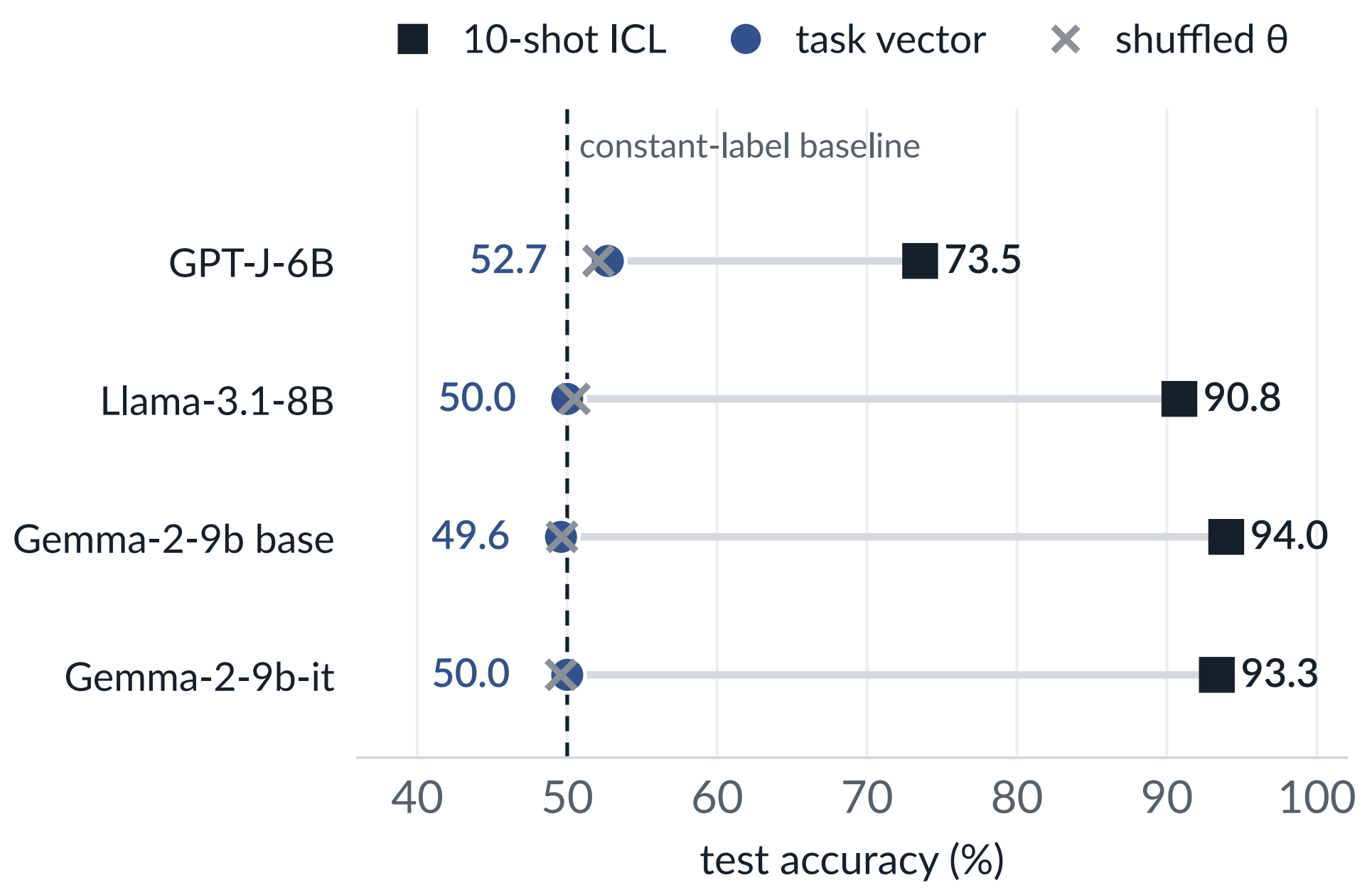


Figure 5: Test accuracy, averaged over both mappings and three seeds.

In 18 of 24 arbitrary-label cells, the patched model gives every input the same label. Natural labels do no better (49.2–50.0%).

What a shuffled θ keeps

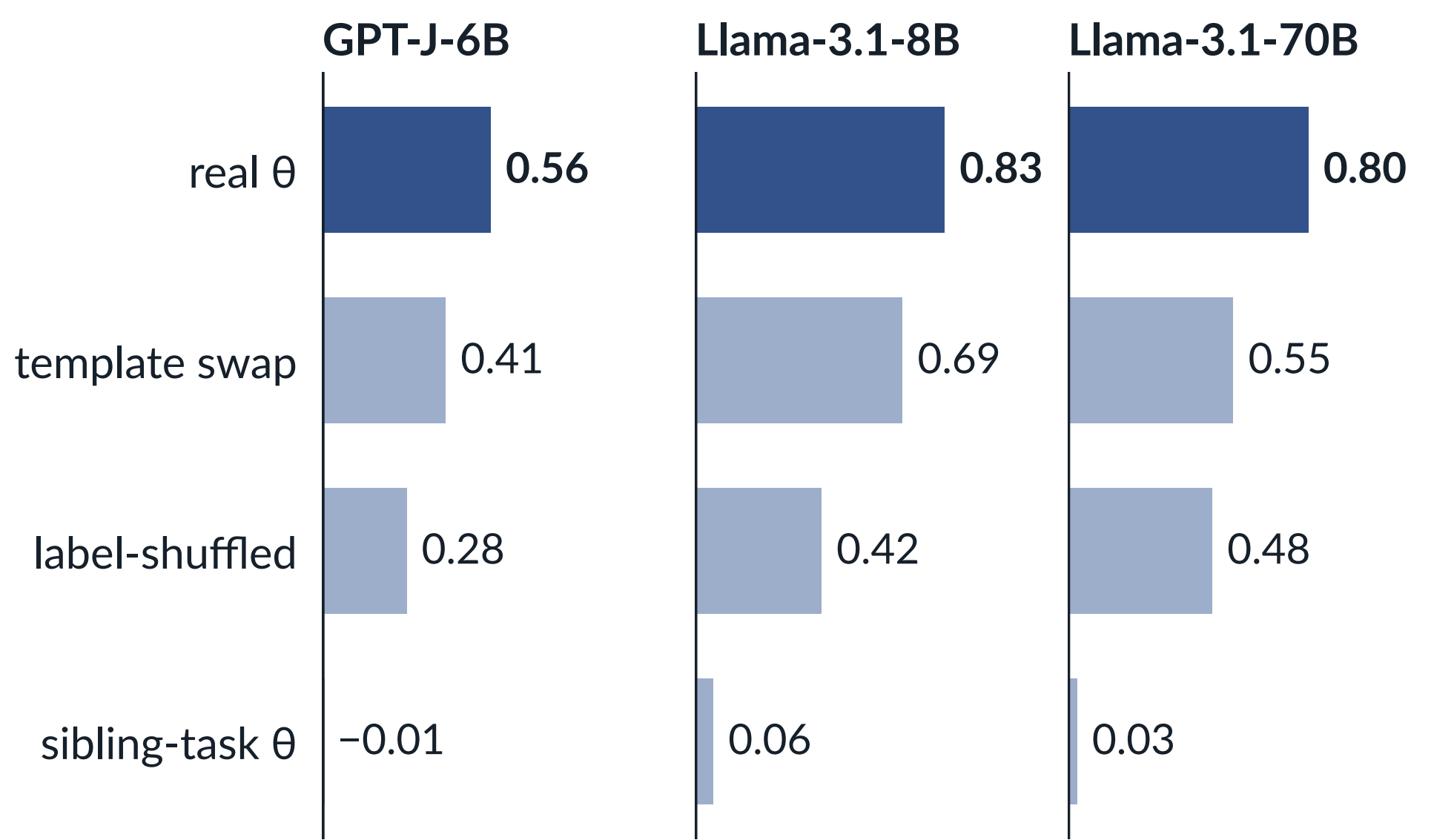


Figure 6: Exploratory recovery at the real TV's layer.

Shuffled-label states still recover 0.28–0.48 of the ICL gap (real: 0.56–0.83); a sibling task's vector recovers almost none. An x -> y extraction template raises GPT-J capitalization from 0.47 to 0.97.

Conclusions

- Success on word pairs does not show broader task transfer. On sentiment, TVs stay near chance on all four checkpoints.
- Controls need their own held-out layer. Gains over zero-shot accuracy can overstate transfer.
- Our FV implementation works on GPT-J and Llama-3.1-8B and fails on both Gemma checkpoints in a two-task follow-up. The cause is open.

References. [1] Todd, Li, Sharma, Mueller, Wallace, Bau. Function vectors in large language models. ICLR 2024. [2] Hendel, Geva, Globerson. In-context learning creates task vectors. Findings of EMNLP 2023. [3] Fiotto-Kaufman et al. NNsight and NDIF: Democratizing access to open-weight foundation model internals. ICLR 2025. [4] Wei et al. Larger language models do in-context learning differently. arXiv:2303.03846, 2023.

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